Towards Transparent and Accurate Diabetes Prediction

Integrating Ensemble Learning and Explainable AI (XAI) for Clinical Trust.

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01 | Problem

? RESEARCH PROBLEM

Traditional AI models are "Black Boxes." A doctor cannot trust a "Yes/No" answer without knowing the logic.

! Key Insight

Diabetes is a global crisis, early, understandable detection can change lives.

02 | Research Gap

? The Limitation

Most current projects focus only on accuracy. They ignore the "Interpretability" (the why).

! The Solution

This project implements an Ensemble Model (for power) + XAI Tools (for transparency).

! Goal

Reach >90% accuracy while providing visual proofs for every prediction

03 | Dataset

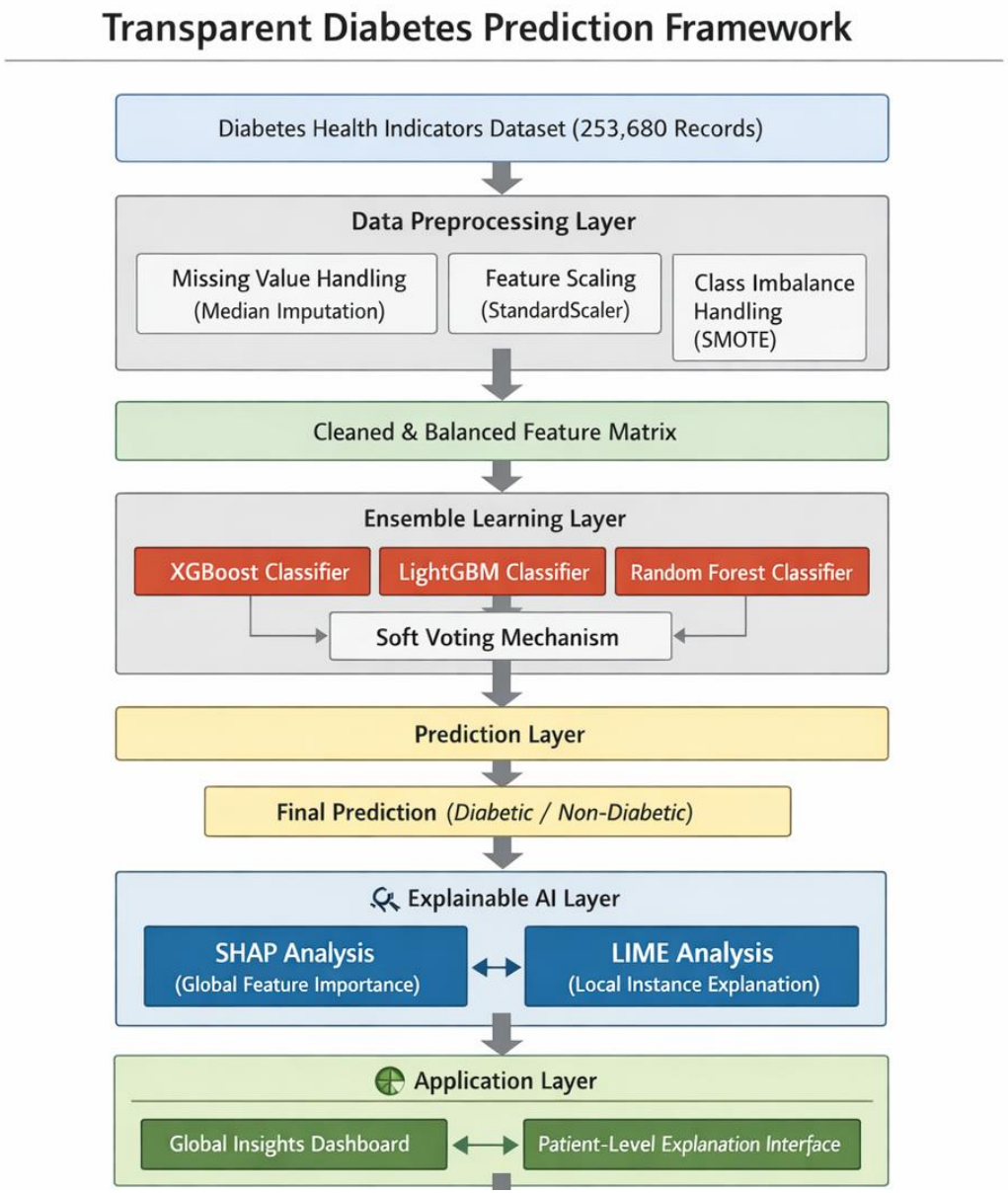
? Source

Diabetes Binary Health Indicators Dataset (253,680 records).

! Features

21 health indicators (High BP, Cholesterol, BMI, Smoker, Stroke, Heart Disease, etc.).

04 | Framework



04 | Methodology

? Handling Imbalance

Using SMOTE (Synthetic Minority Over-sampling Technique). We don't just copy data we create new, logical data points to help the AI learn what "Diabetes" looks like

! Standardization

Using StandardScaler to ensure a "BMI of 30" and "Age of 50" are mathematically comparable.

! Data Cleaning

Handling missing values via Median Imputation to maintain data integrity.

05 | Model Architecture

? The Ensemble Approach

Using a "Soft Voting" classifier (The Decider)

! Models

XGBoost: Excellent for speed and performance.

LightGBM: Optimized for speed and large-scale data efficiency

Random Forest: Reduces the risk of "overfitting" (the AI memorizing instead of learning)

06 | The "Glass Box" – Explainable AI (XAI)

? SHAP (Shapley Additive Explanations)

Used for Global understanding. It tells us: "Across the whole country, what are the top 3 causes of diabetes?" (Result: BMI, Age, General Health).

! LIME (Local Interpretable Model-agnostic Explanations)

Used for Individual understanding. It tells the doctor: "Patient X is at risk specifically because of their high Blood Pressure, even though their BMI is normal."

! The 90/90 Target

Achieving >90% Accuracy and >90% Interpretability (measured by SHAP consistency).

! Explainable Dashboard

A prototype interface showing "Local Explanations" (why this patient is sick) and "Global Explanations" (what causes diabetes in the general population).

Thank You